Equivariant Reinforcement Learning Policies with Symmetry Breaking

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Background and Motivation. Modern deep reinforcement learning (RL) provides promising solutions to sequential decision making tasks such as robotics, optimization, and controls problems; however they suffer from high sample complexity and lack of generalization. In deep learning, *Group Equivariant Neural Networks* leverage symmetries present in a problem space to improve sample efficiency and generalization by enforcing them as inductive biases [1] [2]. For example, symmetries arise in physics (e.g. conservation laws) [3] as well as robotics and RL tasks (optimal policies are often invariant to flips, translations, and rotations, for example Figure 1) [6] [7]. However, while these symmetries are often present in RL tasks, in practice they are almost always approximate or even broken explicitly, thus rendering equivariant learning ineffective [5] [9].

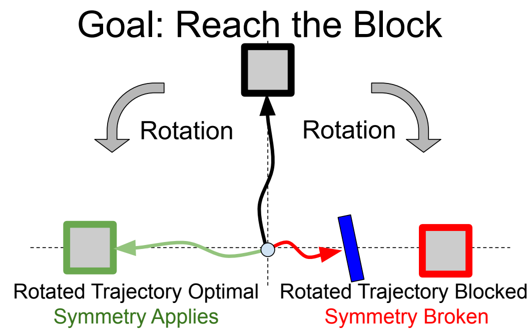

Figure 1: Equivariant policy and symmetry breaking. The task (reach the block) is equivariant under a counterclockwise rotation, but symmetry is broken under a clockwise rotation because of an obstacle.

Methods and Objectives. My Masters research project will address this key challenge by developing equivariant RL algorithms that are sample efficient, generalize well, and are robust to symmetry breaking. To do so, I will combine techniques from mathematical analysis, geometric learning, RL, and neural network design to pursue the following research objectives. Together, these three objectives will help bring the data efficiency of geometric deep learning to real world problems by relaxing their overly-strict assumptions of hard symmetries.

(O1) *Detecting local explicit symmetry breaking*. To effectively leverage symmetries in learning algorithms, we first must characterize where those symmetries apply in practice. My first objective will be to develop an algorithm using tools from hypothesis testing which can detect where within a state space those symmetries are broken in a robotics trajectory dataset [4].

(O2) *Developing locally equivariant NN architectures*. To capitalize on the knowledge of the overarching symmetry and where and how it is broken, the RL policies must be designed to obey them as inductive biases. Thus, my next objective will be to develop NN architectures that appropriately relax the equivariance constraints in equivariant neural networks so that the symmetries hold and can be broken in appropriate regions of the state space [8] [9].

(O3) *Equivariant RL for robotics*. Current work on equivariant RL primarily focuses on simple environments where task symmetries hold exactly, leading to promising but limited demonstrations. My last objective will be to investigate cases when locally equivariant models can outperform baselines in real robotic settings, where symmetries are only approximate due to robot kinematics, obstacles, and physical interactions. I will evaluate my policies in simulation using MuJoCo-based benchmarks against non-equivariant baselines, and on real robots in our lab.

Work Plan (Award Period). Throughout the course of my masters, I will work on (O1) in months 1-5, and (O2) in months 6-12, and I will seek to work on (O3) concurrently, but finish testing in months 13-18.

Significance and Expected Contributions. Unlike the internet scale-data used to train large-scale language and vision transformer models, the trajectory data used to train robotics is scarce. Furthermore, robot learning problems are complex, with slight modifications to tasks potentially resulting in catastrophic failures. The techniques developed through this masters will address both the sample efficiency and brittleness problems that plague modern RL and AI methods, enabling their application in robotics.

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